Lecture 1 - Programming Paradigm & POP

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What is Programming Paradigm?

Programming paradigm is a style of writing program codes.
There are various ways of coding with respect to the needs. Likewise we have different programming languages for each of them.

There are different types of programming paradigm like :-

1. Procedure Oriented Programming - C, Pascal, FORTRAN, COBOL
2. Object Oriented Programming - Java, C++, Python
3. Logic Programming - Prolog, Datalog
4. Functional Programming - Haskell, Scalla, Lisp
5. Event Driven Programming - JavaScript, Visual Basic, C#

Procedure Oriented Programming (POP)

In POP, a program is divided into functions (procedures). The main focus is on what to do — i.e., the sequence of steps/instructions

Key Characteristics of POP:

- Program is divided into functions
- Data moves freely between functions — less secure
- Top-down approach: problem broken into smaller sub-problems
- Functions can access and modify global data — risky in large programs

Advantages

- Simple and Easy to Understand
- Easy to Divide Programs into Functions
- Faster Program Development for Small Problems
- Functions Can be Reused
- Require Less Memory
- Suitable for Mathematical and Sequential Problems - Calculations, algorithms, data processing

Disadvantages

- Poor data security
- Difficult to manage large programs
- No data hiding
- Harder to maintain
- Less real world modelling
- Code reusability is limited

Example Code:

```c
#include <stdio.h>

// Global data
int totalBill = 0;

// Function to add pizza price
void orderPizza()
{
    printf("Pizza ordered - Rs.300\n");
    totalBill = totalBill + 300;
}

// Function to add burger price
void orderBurger()
{
    printf("Burger ordered - Rs.150\n");
    totalBill = totalBill + 150;
}

// Function to display final bill
void displayBill()
{
    printf("Total Bill = Rs.%d\n", totalBill);
}

// Main function
int main()
{
    printf("Welcome to the Restaurant\n\n");
    orderPizza();
    orderBurger();
    displayBill();
    return 0;
}
```